

# The role of expert judgement in insurance internal model for operational risk

Roberto Baviera<sup>1</sup>, Pietro Manzoni<sup>2</sup>, Michele D. Massaria<sup>1</sup>

May 16, 2026

<sup>1</sup> Politecnico di Milano, Department of Mathematics, Italy.

<sup>2</sup> University of Edinburgh, Business School, United Kingdom.

(Preliminary version, do not distribute)

## Abstract

Estimating correlations in operational risk is a core requirement for internal models under the Solvency II framework, but the sparse nature of operational losses makes it difficult to apply standard methods effectively. In this paper, we introduce a parsimonious semi-parametric Gaussian copula-based model, and we present efficient algorithms to estimate correlations and simulate losses. We apply the model to operational loss data from GGG, the largest operational risk database that includes all major insurance companies worldwide. Even considering all losses observed in the main companies across all Western countries over a long time interval, we find that correlations cannot be estimated with sufficient accuracy for regulatory capital purposes. A clear policy implication follows from this study: it highlights the key role of expert judgment in internal models for operational risk, as statistical methods alone cannot deliver the accuracy required under Solvency II.

**Keywords:** Operational risk, semi-parametric copula model, expert judgement, GGG

## Address for correspondence:

**Roberto Baviera**

Department of Mathematics

Politecnico di Milano

32 p.zza Leonardo da Vinci

I-20133 Milano, Italy

Tel. +39-02-2399 4575

roberto.baviera@polimi.it

# 1 Introduction

Internal models play a crucial role in the financial industry, enabling institutions to assess and manage risk effectively. In the context of operational risk, these models aim to quantify potential losses by accounting for the dependencies between different categories of risk (see, e.g., Bolancé *et al.* 2012). Correlation is a key element in this process, as it determines how risks interact and aggregate. Accurately assessing the correlations is essential for obtaining reliable estimates of total risk exposure, which in turn directly affects capital allocation and financial stability (see, e.g., Embrechts 2002, McNeil *et al.* 2015).

Unlike market or credit risk, where extensive and granular datasets enable robust statistical estimation, operational risk is characterized by sparse time series (see, e.g., Böcker and Klüppelberg 2010, Bolancé *et al.* 2012). Operational losses are typically infrequent events, resulting in datasets that contain long sequences of zero observations and only occasional positive losses. The scarcity of co-occurring events makes it difficult to estimate dependence parameters with high reliability using purely statistical methods (see, e.g., Yip and Yau 2005, Bandi *et al.* 2020). Consequently, empirical estimates of correlation between risks are often subject to considerable uncertainty, and *expert judgment* becomes essential for selecting appropriate dependence parameters, ensuring that internal models remain both realistic and consistent with industry experience (see, e.g., Kelliher *et al.* 2017, 2020).

In this paper, we develop a new approach for correlation estimation, tailored to the characteristics of operational risk data. Our method is based on a Gaussian copula representation of dependence – the standard tool for aggregating risk components in the insurance and risk management literature (see, e.g., Cherubini *et al.* 2004, McNeil *et al.* 2015). However, estimation and simulation of such multivariate models are straightforward only when the marginal distributions are continuous. The presence of several zeros in the time series – as typical in operational risk data – introduces additional modeling and mathematical challenges that require careful treatment (see, e.g., Baviera *et al.* 2026).

We introduce a semi-parametric framework that captures dependence through a copula structure while avoiding the need for a full specification of the marginal distributions. This extends the seminal work by Genest *et al.* (1995) for the estimation of copula parameters in presence of continuous marginals to settings characterized by sparse observations. Within this framework, we develop two novel estimators for the correlation parameter of the copula. Both estimators are consistent and asymptotically normal, and we derive methods for constructing confidence intervals (CIs) based on both asymptotic theory and statistical bootstrap (see, e.g., Joe 2005). The semi-parametric nature of our approach ensures flexibility by not requiring explicit specification of the marginal loss distributions, making the proposed correlation coefficient robust to misspecification and suitable for a broad range of applications.

The primary motivation for our methodology arises from the estimation of correlations in

the insurance sector within the Solvency II framework (cf. Solvency II Directive, 2009). Under Solvency II, the aggregation of risks relies on a grid of discrete correlation values ranging from -1 to 1, with fixed increments of 0.25 (i.e. correlations are restricted to the values  $\{0.00, \pm 0.25, \pm 0.50, \pm 0.75, \pm 1.00\}$ ). In principle, empirical estimation could be used to select the most appropriate value within this grid; however, the reliability of such estimates in data-scarce environments remains an open question.

To investigate this issue, we apply our estimators to the GGG database, the largest available dataset of operational losses in insurance, covering institutions in Western Europe and North America between 2008 and 2022 (cf. GGG 2026). Despite the size of the dataset, our empirical results show that statistical uncertainty remains substantial. In particular, the width of the 95% CIs for the correlation parameter always exceeds 0.25: as a result, intervals often span multiple values of the Solvency II correlation grid, making it impossible to determine a unique correlation level solely on the basis of statistical evidence. These findings highlight an important practical implication: even when using a flexible and efficient statistical framework, empirical data alone may be insufficient to determine correlation parameters in operational risk models.

The main contributions of this paper are fourfold. First, we develop a new semi-parametric framework for modeling dependence in operational risk and introduce two novel estimators for the correlation parameter. Second, we provide a rigorous asymptotic analysis of the estimators and construct CIs using both asymptotic methods and semi-parametric bootstrap techniques. Third, we conduct a large-scale empirical analysis using the GGG dataset, providing new insights into the dependence structure of operational loss data. Finally, we demonstrate that, even with improved statistical tools, empirical estimates alone may not be sufficient to determine correlations within the Solvency II framework, thereby emphasizing the continued importance of expert judgment in the construction of internal models.

The remainder of the paper is organized as follows. Section 2 presents the semi-parametric model and discusses its likelihood structure and simulation procedure. Section 3 introduces the proposed estimators and develops the methodology for point estimation and CI construction. Finally, Section 4 reports the empirical results obtained from applying the model to the GGG dataset. Finally, Section 5 states our conclusions.

## 2 Model description

We consider a semi-parametric framework. Unlike fully parametric modeling approaches (see, e.g., Baviera *et al.* 2026), where the marginal distributions for non-zero losses are explicitly specified (e.g., as lognormal distributions), a semi-parametric approach considers only

We consider a semi-parametric special case of the general framework proposed by Baviera *et al.* (2026) to model time series of insurance claims. Unlike the fully parametric approach, where the marginal distributions for non-zero losses (denoted as  $\psi_1$  and  $\psi_2$ ) must be explicitly specified

(e.g., as lognormal distributions), here we leave them unspecified. Instead, we focus solely on the parameters governing the occurrence of non-zero losses and their dependence. Thus, the parameter space is reduced to  $[p_1, p_2, \varrho]^\top \in [0, 1] \times [0, 1] \times (-1, 1)$ , where  $p_1$  and  $p_2$  are the marginal probabilities of observing a positive loss in risk classes  $X_1$  and  $X_2$ , respectively, and  $\varrho$  is the correlation parameter of a bivariate Gaussian copula.

The pdf and the cdf of the two marginal distributions ( $i = 1, 2$ ) are

$$\begin{cases} f_i(x; p_i) := (1 - p_i)\delta_0(x) + p_i\psi_i(x_i) \mathbb{1}_{\{x>0\}}, \\ F_i(x; p_i) := (1 - p_i)\mathbb{1}_{\{x\geq 0\}} + p_i\Psi_i(x) \mathbb{1}_{\{x>0\}}. \end{cases} \quad (2.1)$$

The joint cdf of the bivariate random vector  $[X_1, X_2]^\top$  is then constructed via Sklar's theorem (see, e.g., McNeil *et al.* 2005, Th.5.3, p.186):

$$\mathbb{P}(X_1 \leq x_1, X_2 \leq x_2) = C(F_1(x_1; p_1), F_2(x_2; p_2); \varrho), \quad (2.2)$$

where  $C(\cdot, \cdot; \varrho)$  denotes the bivariate Gaussian copula with correlation  $\varrho$ .

Since the marginals have a probability atom at zero, the sample space can be partitioned into four distinct regions: *i*) no losses in either risk class, *ii*) a loss only in the first class, *iii*) a loss only in the second class, and *iv*) simultaneous losses in both classes (co-jumps). In the proposed model, the probabilities associated with these four regions are fully determined by the parameters  $[p_1, p_2, \varrho]^\top$  and are defined as follows:

$$\begin{cases} p_0 := \mathbb{P}(X_1 = 0, X_2 = 0) = C(1 - p_1, 1 - p_2; \varrho) \\ p_1^\perp := \mathbb{P}(X_1 > 0, X_2 = 0) = p_1 - p^\parallel \\ p_2^\perp := \mathbb{P}(X_1 = 0, X_2 > 0) = p_2 - p^\parallel \\ p^\parallel := \mathbb{P}(X_1 > 0, X_2 > 0) = \overline{C}(1 - p_1, 1 - p_2; \varrho) \end{cases} \quad (2.3)$$

where  $\overline{C}$  denotes the survival copula. These probabilities play a pivotal role in our semi-parametric framework, as they will directly constitute the core of the estimation procedure developed in Section 3.

A key analytical property of the model is its tractable likelihood. Despite the presence of a mass at zero in both marginals – which complicates the standard copula likelihood derivation – the log-likelihood admits an explicit closed form, organized by the four regions defined above.

**Proposition 2.1.** *The log-likelihood contribution of a single observation  $[x_1, x_2]^\top$  is*

$$\begin{aligned} \ell(p_1, p_2, \varrho \mid x_1, x_2) = & \ln p_0 \mathbb{1}_{\{x_1=0, x_2=0\}} + \ln(p_1 \psi_1(x_1)) \mathbb{1}_{\{x_1>0\}} + \ln(p_2 \psi_2(x_2)) \mathbb{1}_{\{x_2>0\}} + \\ & + \ln \left( \frac{\partial C(u_1, 1 - p_2; \varrho)}{\partial u_1} \bigg|_{u_1=F_1(x_1; p_1)} \right) \mathbb{1}_{\{x_1>0, x_2=0\}} + \\ & + \ln \left( \frac{\partial C(1 - p_1, u_2; \varrho)}{\partial u_2} \bigg|_{u_2=F_2(x_2; p_2)} \right) \mathbb{1}_{\{x_1=0, x_2>0\}} + \\ & + \ln c(F_1(x_1; p_1), F_2(x_2; p_2); \varrho) \mathbb{1}_{\{x_1>0, x_2>0\}} , \end{aligned} \quad (2.4)$$

where  $c(\cdot, \cdot; \varrho)$  denotes the Gaussian copula density. Given a time series  $\mathcal{X} := \{[x_1^{(t)}, x_2^{(t)}]\}_{t=1}^N$  of length  $N$ , the full sample log-likelihood is

$$\begin{aligned} \mathcal{L}(p_1, p_2, \varrho \mid \mathcal{X}) = & n_0 \ln p_0 + \sum_{t \mid x_1^{(t)} > 0} \ln p_1 \psi_2(x_1^{(t)}) + \sum_{t \mid x_2^{(t)} > 0} \ln p_2 \psi_2(x_2^{(t)}) + \\ & + \sum_{t \mid x_1^{(t)} > 0, x_2^{(t)} = 0} \ln \left( \frac{\partial C(u_1, 1 - p_2; \varrho)}{\partial u_1} \bigg|_{u_1=F_1(x_1^{(t)}; p_1)} \right) + \\ & + \sum_{t \mid x_1^{(t)} = 0, x_2^{(t)} > 0} \ln \left( \frac{\partial C(1 - p_1, u_2; \varrho)}{\partial u_2} \bigg|_{u_2=F_2(x_2^{(t)}; p_2)} \right) + \\ & + \sum_{t \mid x_1^{(t)} > 0, x_2^{(t)} > 0} \ln c(F_1(x_1^{(t)}; p_1), F_2(x_2^{(t)}; p_2); \varrho) . \end{aligned} \quad (2.5)$$

*Proof.* See Appendix B □

This explicit representation of the full sample log-likelihood (2.5) is crucial: differentiating it with respect to the parameters provides the theoretical foundation for the semi-parametric estimating equations developed in Section 3.

A final, crucial feature of the proposed framework is its straightforward simulation. While the explicit representation of the full-sample log-likelihood (2.5) provides the theoretical foundation for point estimation, an efficient simulation scheme is essential for assessing the uncertainty of the estimators via a semi-parametric bootstrap.

The proposed semi-parametric simulation approach leverages the empirical cumulative distribution function (ecdf). The procedure is remarkably simple and consists of two steps. First, we generate a vector  $[u_1^{(t)}, u_2^{(t)}]^\top$  that follows a bivariate Gaussian copula with correlation coefficient  $\varrho$ . Then, we apply a threshold mechanism: for each risk class  $i$ , if  $u_i^{(t)} > 1 - p_i$ , the value  $u_i^{(t)}$  is retained as the simulated ecdf for that observation. Conversely, if  $u_i^{(t)} \leq 1 - p_i$ , it is assigned a null value to represent a period without losses.

At the end of the semi-parametric simulation, the empirical probabilities of each of the four

regions – i.e. the empirical counterparts of the probabilities in (2.3) – are obtained by counting the four possible types of events: both  $u_1, u_2$  are zero; a positive  $u_1$  for  $X_1$ , zero for  $X_2$ ; a positive  $u_2$  for  $X_2$ , zero for  $X_1$ ; both losses (and thus  $u_1, u_2$ ) are positive.

#### Algorithm 1. Semi-Parametric Simulation

**Require:**  $N$  (number of simulations)

```

1: for  $t = 1, \dots, N$  do
2:   Draw  $[\xi_1^{(t)}, \xi_2^{(t)}]^\top$  from st.n. with correlation  $\varrho$            ▷ Step (i): Gaussian copula
3:   for  $i \in \{1, 2\}$  do
4:     Define  $u_i^{(t)} := \Phi(\xi_i^{(t)})$ 
5:     if  $u_i^{(t)} \leq 1 - p_i$  then                                     ▷ Step (ii): Thresholding rule
6:       Set  $u_i^{(t)} \leftarrow 0$ 

```

## 3 Correlation Estimators

In this Section, we introduce two novel semi-parametric estimators for the copula correlation parameter.

We adopt a two-stage estimation procedure that is an application of the Inference Functions for Margins (IFM) in a semi-parametric setting. In our framework, the first stage determines the probabilities of observing a positive loss ( $p_1$  and  $p_2$ ) and, when needed, the empirical cumulative distribution functions of the strictly positive losses. In the second stage, the copula correlation  $\varrho$  is estimated conditioned on these marginal quantities. This method is an example of cascade estimation, where the parameters related to model parts requiring greater accuracy are calibrated first, with the remaining parameters estimated subsequently (see, e.g. Brigo and Morini 2005, Baviera and Massaria 2026).

Both estimators are formally derived within the broad framework of estimating functions theory, briefly described in Appendix A. This theoretical foundation guarantees that the proposed estimators are consistent and asymptotically normal, allowing for the rigorous construction of confidence intervals. In the remainder of this Section, we define the estimating equations for each method and evaluate their specific asymptotic properties.

### 3.1 Semi-parametric estimators: Estimator 1

The first semi-parametric estimator we propose relies strictly on the observed frequencies of the four disjoint regions defined in equation (2.3). Its core advantage lies in its extreme simplicity: it does not depend on the specific values or empirical distributions of the non-null losses, but only on the joint occurrence of the events.

### 3.1.1 Point Estimate

In this semi-parametric copula setting, we have to estimate three parameters  $[p_1, p_2, \varrho]^\top$ , which can be obtained from the following vector-valued estimating function:

$$\begin{cases} g_1(p_1, p_2, \varrho) := \mathbb{1}_{\{X_1 > 0\}} - p_1 \\ g_2(p_1, p_2, \varrho) := \mathbb{1}_{\{X_2 > 0\}} - p_2 \\ g_3(p_1, p_2, \varrho) := \mathbb{1}_{\{X_1=0, X_2=0\}} - p_0 = \mathbb{1}_{\{X_1=0, X_2=0\}} - C(1 - p_1, 1 - p_2; \varrho) \end{cases} . \quad (3.1)$$

To obtain the parameter estimates, we equate the empirical sum of these functions over the sample to zero, as per equation (A.2). Let  $N$  be the total length of the time series, and let  $n_1$  and  $n_2$  be the total number of strictly positive losses observed in risk classes  $X_1$  and  $X_2$ , respectively. We further define  $n_0$  as the number of simultaneous zero losses (no-jumps), and  $n^\parallel$  as the number of simultaneous positive losses (co-jumps) observed across both classes.

**Proposition 3.1.** *For the semi-parametric estimator, the vector of estimators is:*

$$\begin{cases} \hat{p}_1 = \frac{n_1}{N} , \\ \hat{p}_2 = \frac{n_2}{N} , \\ \hat{\varrho} \text{ s.t. } C(1 - \hat{p}_1, 1 - \hat{p}_2; \hat{\varrho}) = \frac{n_0}{N} . \end{cases} \quad (3.2)$$

*Proof.* See Appendix B □

The strength of Estimator 1 lies in its simplicity: the parameters  $p_1, p_2, \varrho$  are calibrated by directly matching the observed frequencies to their corresponding theoretical probabilities.<sup>1</sup>

### 3.1.2 Confidence Intervals

The Confidence Intervals can be constructed in two ways: via semi-parametric bootstrap, and via the theory of estimating functions. In the following, we present and describe the two approaches.

The semi-parametric bootstrap is straightforward to implement thanks to the efficient simulation Algorithm 1 described in Section 2. The procedure involves three steps. First, we simulate  $B$  replicas of the bivariate time series using the point estimates  $[\hat{p}_x, \hat{p}_y, \hat{\varrho}]^\top$ . Second, for each

---

<sup>1</sup>Let us observe that it is mathematically equivalent to estimate the correlation  $\varrho$  using the frequency of the no-jumps or that of the co-jumps. Indeed, the equation  $g_3(p_1, p_2, \varrho)$  can be equivalently substituted with

$$g_4(p_1, p_2, \varrho) := \mathbb{1}_{\{X_1 > 0, X_2 > 0\}} - p^\parallel ,$$

in such case, the estimator  $\hat{\varrho}$  satisfies  $\overline{C}(1 - \hat{p}_1, 1 - \hat{p}_2; \hat{\varrho}) = \frac{n^\parallel}{N}$ .

simulated replica, we count the number of no-jumps  $n_0$  and invert equation (3.2) to obtain a bootstrapped estimate of  $\hat{\varrho}$ . Finally, we construct the confidence interval by deriving the empirical quantiles of order  $\alpha/2$  and  $1 - \alpha/2$  from the bootstrapped distribution (e.g.,  $\alpha = 0.05$  for a 95% CI).

Alternatively, asymptotic confidence intervals can be obtained analytically by leveraging the theory of estimating functions (see Appendix A). The exact asymptotic distribution of the estimator is formally described in the following Proposition.

**Proposition 3.2.** *The asymptotic distribution of the estimators  $[\hat{p}_1, \hat{p}_2, \hat{\varrho}]^\top$ , obtained via (3.1), is*

$$\begin{bmatrix} \hat{p}_1 \\ \hat{p}_2 \\ \hat{\varrho} \end{bmatrix} \sim \mathcal{N} \left( \begin{bmatrix} p_1 \\ p_2 \\ \varrho \end{bmatrix}, \frac{1}{N} D^{-1} M (D^{-1})^\top \right)$$

where  $N$  is the length of the time-series and

$$M := \begin{bmatrix} p_1(1-p_1) & p^\parallel - p_1p_2 & -p_1p_0 \\ p^\parallel - p_1p_2 & p_2(1-p_2) & -p_2p_0 \\ -p_1p_0 & -p_2p_0 & p_0(1-p_0) \end{bmatrix} \quad (3.3)$$

$$D := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ d_{31} & d_{32} & d_{33} \end{bmatrix} \quad \text{where} \quad \begin{cases} d_{31} := \Phi \left( \frac{1}{\sqrt{1-\varrho^2}} [\Phi^{-1}(1-p_2) - \varrho\Phi^{-1}(1-p_1)] \right) \\ d_{32} := \Phi \left( \frac{1}{\sqrt{1-\varrho^2}} [\Phi^{-1}(1-p_1) - \varrho\Phi^{-1}(1-p_2)] \right) \\ d_{33} := \varphi_2(\Phi^{-1}(1-p_1), \Phi^{-1}(1-p_2); \varrho) \end{cases} \quad (3.4)$$

*Proof.* See Appendix B □

It is immediate to derive the asymptotic distribution of the correlation estimator.

**Corollary 3.3.** *The correlation  $\hat{\varrho}$  obtained via Estimator 1 is asymptotically distributed as a Gaussian rv*

$$\hat{\varrho} \sim \mathcal{N} \left( \varrho, \frac{1}{N} v^\top M v \right) \quad (3.5)$$

where  $M$  is defined in (3.3) and  $v := [-d_{31}/d_{33} \quad -d_{32}/d_{33} \quad -1/d_{33}]^\top$  is the transpose of the last row of the inverse of the matrix  $D$  in (3.4).

*Proof.* See Appendix B □

The Corollary allows us to determine the asymptotic confidence intervals for the estimator of the correlation  $\varrho$ . As standard in statistical inference theory, the exact expression for the asymptotic variance is approximated using the estimate  $[\hat{p}_1, \hat{p}_2, \hat{\varrho}]^\top$ , since the true triplet of parameters  $[p_1, p_2, \varrho]^\top$  is unknown (see, e.g., Casella and Berger 2024).

Estimator 1 offers a straightforward approach to estimate the correlation without the need for specifying the form of the marginals, and shows to be particularly adapt to time-series which are rather (but not totally) empty. The main limitation of this estimator is that the confidence intervals have large confidence intervals for  $p_1, p_2$  close to 1, and the estimator becomes undefined in the limit  $p_1, p_2 \rightarrow 1$ .

For this reason in the next Section, we introduce an improved semi-parametric estimator, designed to exploit at best the information in the time-series. It considers not only the number of events in each of the four regions, but also the size of the observed losses. This estimator appears to be (slightly) more precise in all cases, at a cost of computational cost; we will also prove that this estimator for correlation is well defined in the limit  $p_1, p_2 \rightarrow 1$ .

## 3.2 Semi-parametric estimators: Estimator 2

The second semi-parametric estimator we consider is obtained by replacing the parametric cdf of the marginals with their empirical counterpart. This approach for copula calibration dates back to the works by Oakes (1994) and Genest *et al.* (1995).

In a semi-parametric model, the marginal distributions of positive losses do not have a predefined functional form – governed by some set of distributional parameters – but are reconstructed as empirical cdfs and pdfs. Consequently, unlike fully parametric models, this approach does not require estimating distributional parameters for the severities. The only parameters that need to be estimated are  $p_1$  and  $p_2$  (which model the probability of non-null losses) and the dependence parameter  $\varrho$ , while the marginals are entirely handled via their empirical distributions

The estimator 2, which is presented in this Section, does not depend on the marginal distributions  $\psi_i(x_i; \theta_i)$ ; it just considers the number of co-jumps ( $n^{\parallel}$ ), no-jumps ( $n_0$ ), and the empirical distributions of the losses (but not the losses themselves).

### 3.2.1 Point Estimate

Even in this case, we exploit the theory of estimating function and we calibrate the parameters following a cascade calibration procedure. The parameters to estimate are  $[p_1, p_2, \varrho]$  and the considered vector of estimating equations is:

$$\begin{cases} g_1(p_1, p_2, \varrho) := \mathbb{1}_{\{X_1 \neq 0\}} - p_1 \\ g_2(p_1, p_2, \varrho) := \mathbb{1}_{\{X_2 \neq 0\}} - p_2 \\ g_3(p_1, p_2, \varrho) := g_{\varrho}(p_1, p_2, \varrho; u_1, u_2) \end{cases} \quad (3.6)$$

where

$$\begin{aligned}
g_\varrho(p_1, p_2, \varrho; u_1, u_2) := & \frac{\varphi_2(\Phi^{-1}(1-p_1), \Phi^{-1}(1-p_2); \varrho)}{\Phi_2(\Phi^{-1}(1-p_1), \Phi^{-1}(1-p_2); \varrho)} \mathbb{1}_{\{u_1 \leq 1-p_1, u_2 \leq 1-p_2\}} + \\
& + \frac{\varrho\Phi^{-1}(1-p_2) - \Phi^{-1}(u_1)}{(1-\varrho^2)^{3/2}} \mathcal{M}\left(-\frac{\Phi^{-1}(1-p_2) - \varrho\Phi^{-1}(u_1)}{\sqrt{1-\varrho^2}}\right) \mathbb{1}_{\{u_1 > 1-p_1, u_2 \leq 1-p_2\}} + \\
& + \frac{\varrho\Phi^{-1}(1-p_1) - \Phi^{-1}(u_2)}{(1-\varrho^2)^{3/2}} \mathcal{M}\left(-\frac{\Phi^{-1}(1-p_1) - \varrho\Phi^{-1}(u_2)}{\sqrt{1-\varrho^2}}\right) \mathbb{1}_{\{u_1 \leq 1-p_1, u_2 > 1-p_2\}} + \\
& + \left( \frac{\varrho}{1-\varrho^2} - \frac{\varrho}{(1-\varrho^2)^2} [\Phi^{-1}(u_1)^2 + \Phi^{-1}(u_2)^2] + \right. \\
& \quad \left. + \frac{1+\varrho^2}{(1-\varrho^2)^2} \Phi^{-1}(u_1)\Phi^{-1}(u_2) \right) \mathbb{1}_{\{u_1 > 1-p_1, u_2 > 1-p_2\}} .
\end{aligned} \tag{3.7}$$

where  $\mathcal{M}(x) := \varphi(x)/\Phi(-x)$  denotes the inverse Mills ratio (see, e.g., Grimmett and Stirzaker 2020).

In the estimating function  $g_\varrho(p_1, p_2, \varrho; u_1, u_2)$ , we need to classify each observation in the time-series in the four regions ( $X_1 = 0, X_2 = 0$ , no-jump;  $u_1 > 1-p_1, X_2 = 0$ , only  $X_2$  loss;  $X_1 = 0, u_2 > 1-p_2$ , only  $X_1$  loss;  $u_1 > 1-p_1, u_2 > 1-p_2$ , co-jump), while  $u_1$  and  $u_2$  are the empirical estimations of the marginals  $F_x$  and  $F_y$ , respectively. These are the all pieces of information available in a semi-parametric description.

It is useful to comment on the fact that the vector of functions in (3.6) satisfies the definition of estimating function (see Definition A.1, Appendix A). Even in this case,  $d_{33}$  turns out to never be equal to zero, as extensive numerical experiments show. Hence,  $D$  is non-singular.

The matrix  $M$  can be singular when one parameter is known without uncertainty. Two are the relevant examples. When  $p_1 = 0$  (i.e. it sure to have losses equal to zero for the first risk class  $x$ ), then  $p^\parallel = 0$ , and  $M$  has the first row identically equal to zero.<sup>2</sup> If both  $p_1, p_2 = 1$  (i.e. if both time series have no losses equal to zero), then the first two rows of  $M$  become

$$\begin{bmatrix} 0 & 0 & m_{13} \end{bmatrix}$$

i.e. linearly dependent and the matrix  $M$  is singular. In both cases, it is possible to reduce the problem with one with fewer parameters that is well defined.

---

<sup>2</sup>Equivalent results hold when  $p_2 = 0$ , with the second row identically equal to zero.

**Proposition 3.4.** *For the semi-parametric Estimator 2, the vector of estimators is:*

$$\begin{cases} \hat{p}_x = \frac{n_x}{N} \\ \hat{p}_y = \frac{n_y}{N} \\ \varrho \quad \text{s.t.} \quad 0 = h^{p_1, p_2}(\varrho) := \sum_{t=1}^N g_{\varrho} \left( \hat{p}_1, \hat{p}_2, \varrho; u_1^{(t)}, u_2^{(t)} \right) \end{cases} \quad (3.8)$$

*Proof.* See Appendix B □

First, the marginal parameters are obtained from the observed frequencies of non-null values in the time-series. Then, given  $\hat{p}_x, \hat{p}_y$ , the last estimating equation is solved. It plays a key role in the point estimation of the correlation (on the “original” time-series of losses) and in the determination of the confidence interval (via the semi-parametric bootstrap). When estimating the parameters on the “original” time-series,  $u_1^{(t)}$  and  $u_2^{(t)}$  are the empirical estimations of the marginals  $F_1(x_1^{(t)}; p_1)$  and  $F_2(x_2^{(t)}; p_2)$ , respectively. The last estimating equation is also considered on the “simulated” time-series of the semi-parametric bootstrap; in this case, the semi-parametric simulation returns directly the values for  $u_1$  and  $u_2$ , where a zero conventional value is associated to the case of null loss (it is useful to remind that the semi-parametric simulation does not return the losses).

### 3.2.2 Confidence Intervals

Even for this other semi-parametric estimator, the Confidence Intervals can be constructed in two ways: via semi-parametric bootstrap, and via estimating functions’ theory. In the following we present and describe the two approaches. The semi-parametric bootstrap is exactly the one described in Section 3.1.2.

Moreover, thanks to the asymptotic normality of the estimator (see Appendix A), we can deduce a relevant property of the considered estimator.

**Proposition 3.5.** *The asymptotic distribution of the estimators  $[\hat{p}_1, \hat{p}_2, \hat{\varrho}]^\top$ , obtained via (3.6), is*

$$\begin{bmatrix} \hat{p}_1 \\ \hat{p}_2 \\ \hat{\varrho} \end{bmatrix} \sim \mathcal{N} \left( \begin{bmatrix} p_1 \\ p_2 \\ \varrho \end{bmatrix}, \frac{1}{N} D^{-1} M (D^{-1})^\top \right)$$

where  $N$  is the length of the time-series and

$$M := \begin{bmatrix} p_1(1-p_1) & p^\parallel - p_1 p_2 & m_{13} \\ p^\parallel - p_1 p_2 & p_2(1-p_2) & m_{23} \\ m_{13} & m_{23} & m_{33} \end{bmatrix} \quad \text{where} \quad \begin{cases} m_{13} := \mathbb{E}[(\mathbb{1}_{\{X_1 > 0\}} - p_1) h_{\varrho}] \\ m_{23} := \mathbb{E}[(\mathbb{1}_{\{X_2 > 0\}} - p_2) h_{\varrho}] \\ m_{33} := \mathbb{E}[(h_{\varrho})^2] - \mathbb{E}[h_{\varrho}]^2 \end{cases} \quad (3.9)$$

with

$$h_{\varrho}(p_1, p_2, \varrho; F_1(X_1; p_1), F_2(X_2; p_2)) = g_{\varrho}(p_1, p_2, \varrho; F_1(X_1; p_1), F_2(X_2; p_2)) + W_1 + W_2 \quad (3.10)$$

and

$$\begin{cases} W_1 &:= \int_0^1 \int_0^1 (\mathbb{1}_{F_1(X_1; p_1) \leq u_1} - u_1) \frac{\partial g_{\varrho}(p_1, p_2, \varrho; u_1, u_2)}{\partial u_1} c(u_1, u_2; \varrho) du_1 du_2 \\ W_2 &:= \int_0^1 \int_0^1 (\mathbb{1}_{F_2(X_2; p_2) \leq u_2} - u_2) \frac{\partial g_{\varrho}(p_1, p_2, \varrho; u_1, u_2)}{\partial u_2} c(u_1, u_2; \varrho) du_1 du_2 \end{cases}; \quad (3.11)$$

and

$$D := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ d_{31} & d_{32} & d_{33} \end{bmatrix} \quad \text{where} \quad \begin{cases} d_{31} := -\mathbb{E} \left[ \frac{\partial g_{\varrho}}{\partial p_1} \right] \\ d_{32} := -\mathbb{E} \left[ \frac{\partial g_{\varrho}}{\partial p_2} \right] \\ d_{33} := -\mathbb{E} \left[ \frac{\partial g_{\varrho}}{\partial \varrho} \right] \end{cases} \quad (3.12)$$

*Proof.* See Appendix B □

And a Corollary similar to the one of Estimator 1 holds true.

**Corollary 3.6.** *The correlation  $\hat{\varrho}$  obtained via Estimator 2 is asymptotically distributed as a Gaussian rv*

$$\hat{\varrho} \sim \mathcal{N} \left( \varrho, \frac{1}{N} v^{\top} M v \right) \quad (3.13)$$

where  $M$  is defined in (3.9) and  $v := \begin{bmatrix} -d_{31}/d_{33} & -d_{32}/d_{33} & -1/d_{33} \end{bmatrix}^{\top}$  is the transpose of the last row of the inverse of the matrix  $D$  defined in (3.12).

*Proof.* See Appendix B □

### 3.2.3 Estimator 2 in absence of null losses

A remarkable property of Estimator 2 is its asymptotic behavior when data sparsity vanishes, i.e. when both probabilities  $p_1$  and  $p_2$  approach 1. In this limit, the semi-parametric estimating equation is equal to the classical log-likelihood of a Gaussian copula model. This result is described in the following Proposition.

**Proposition 3.7.** *The log-likelihood function (3.8) converges to the function  $\bar{h}(\varrho)$*

$$\bar{h}(\varrho) := \frac{N\varrho}{1-\varrho^2} - \frac{1}{(1-\varrho^2)^2} \sum_{t=1}^N \left[ (1+\varrho^2) \Phi^{-1}(u_1^{(t)}) \Phi^{-1}(u_2^{(t)}) - \varrho \left( \Phi^{-1}(u_1^{(t)})^2 + \Phi^{-1}(u_2^{(t)})^2 \right) \right]$$

almost surely, for  $p_1, p_2 \rightarrow 1$ , for any  $\varrho \in (-1, 1)$ .

*Proof.* See Appendix B □

Consequently, in the absence of probability masses at zero, the estimator 2 coincides with the standard IFM estimator of Gaussian copula correlation, as described in the next Theorem.

**Theorem 3.8.** *The estimator  $\hat{\varrho}$ , solution of (3.8), converges almost surely to  $\bar{\varrho}$ , defined as*

$$\bar{h}(\bar{\varrho}) = 0 .$$

*Proof.* See Appendix B □

As a direct consequence of Theorem 3.8, it is important to emphasize that Estimator 2 remains well-defined even in the limiting case of “dense” time series (i.e., as  $p_1, p_2 \rightarrow 1$ ) and maintains robust estimation performance even when there is a scarcity of zeros within the series. This property represents a crucial theoretical and practical advantage over Estimator 1. The latter, by contrast, is not mathematically well-defined in the limit as the marginal probabilities tend to 1 (indicating a complete absence of a point mass at zero). Furthermore, it is prone to large estimation errors, and it yields particularly wide confidence intervals, when the time series exhibit a severely reduced number of zero observations.

## 4 Results

In this Section, we present the empirical application of our semi-parametric framework to real-world operational risk data. In Section 4.1 we describe the data used in this analysis. In Section 4.2, we report the calibration results for both proposed estimators, demonstrating that the statistical uncertainty remains remarkably large. Finally, Section 4.3 generalizes these empirical findings through a theoretical analysis of the asymptotic confidence intervals.

### 4.1 Dataset

We utilize data from the GGG dataset, the most comprehensive global repository of operational risk losses in the insurance sector (cf. GGG 2026). We analyse the time-series for three risks classes: R1, R2, and R3. We consider monthly losses in the time-window 2008-2022; the dataset consists of 180 monthly observations for all large insurance companies in Western Europe and North America.

Table 1 reports the descriptive statistics for the three selected risk categories, including the average, minimum, and maximum observed positive losses, alongside the proportion of non-zero entries in each time series. Despite the extensive time window and the breadth of the underlying database, a significant proportion of the monthly observations records zero losses (with non-zero

| <b>Risk</b> | <b>Observations</b> | <b>Average</b> | <b>Min</b>      | <b>Max</b> | <b>Non-zero entries</b> |
|-------------|---------------------|----------------|-----------------|------------|-------------------------|
| <b>R1</b>   | 131                 | 1.9 Mln€       | 59.1 <i>K</i> € | 34.6 Mln€  | 0.728                   |
| <b>R2</b>   | 96                  | 0.9 Mln€       | 54.5 <i>K</i> € | 19.3 Mln€  | 0.533                   |
| <b>R3</b>   | 51                  | 1.4 Mln€       | 50.5 <i>K</i> € | 12.8 Mln€  | 0.283                   |

Table 1: Descriptive statistics for the three operational risk categories (R1, R2, and R3). We report the average, minimum, and maximum observed positive losses, alongside the proportion of non-zero entries (i.e., months with at least one recorded loss) over the 180-month period.

proportions of 0.728, 0.533, and 0.283 for R1, R2, and R3 respectively). This highlights the fundamental sparse nature of operational risk data.

Furthermore, Table 2 reports, for each pair of risk classes, reporting the number of simultaneous no-jumps (months where neither class recorded a loss) and co-jumps (months where both classes recorded a positive loss).

|           | <b>No-jumps</b> |           |           | <b>Co-jumps</b> |           |           |
|-----------|-----------------|-----------|-----------|-----------------|-----------|-----------|
|           | <b>R1</b>       | <b>R2</b> | <b>R3</b> | <b>R1</b>       | <b>R2</b> | <b>R3</b> |
| <b>R1</b> |                 | 80        | 43        |                 | 33        | 41        |
| <b>R2</b> |                 |           | 27        |                 |           | 60        |
| <b>R3</b> |                 |           |           |                 |           |           |

Table 2: Joint occurrence of events for each pair of risk categories. The table reports the number of simultaneous no-jumps (months with zero losses in both classes) and co-jumps (months with strictly positive losses in both classes).

## 4.2 Correlation estimation

In this Section, we present the empirical results obtained by calibrating our two semi-parametric correlation estimators on the GGG dataset. The primary objective is to assess whether the statistical evidence alone is sufficient to pinpoint a robust correlation value for internal models, particularly in relation to the Solvency II standard framework, which typically operates on a discrete correlation grid with increments of 0.25.

We first evaluate the performance of Estimator 1, which relies exclusively on the joint occurrence frequencies of the loss events. Table 3 presents the point estimates for the Gaussian copula correlation across the three pairs of risk categories. Subsequently, Table 4 reports the corresponding 95% confidence intervals, comparing the results obtained via the semi-parametric bootstrap ( $B = 10^3$  simulations) with those derived analytically from the asymptotic estimating functions theory. The two methodologies yield remarkably consistent bounds. However, a critical observation emerges: the statistical uncertainty is substantial. For all three pairs of risk classes, the width of the 95% CI

strictly exceeds 0.40. Consequently, these intervals always encompass at least two discrete values from the standard regulatory correlation grid.

|           | <b>R1</b> | <b>R2</b> | <b>R3</b> |
|-----------|-----------|-----------|-----------|
| <b>R1</b> |           | 0.414     | 0.306     |
| <b>R2</b> |           |           | −0.008    |
| <b>R3</b> |           |           |           |

Table 3: Point estimates of the correlation parameter  $\hat{\varrho}$  obtained via Estimator 1 for each pair of operational risk categories.

|           | <b>Bootstrap</b> |                |                 | <b>Asymptotic</b> |                |                 |
|-----------|------------------|----------------|-----------------|-------------------|----------------|-----------------|
|           | <b>R1</b>        | <b>R2</b>      | <b>R3</b>       | <b>R1</b>         | <b>R2</b>      | <b>R3</b>       |
| <b>R1</b> |                  | [0.198, 0.610] | [0.058, 0.569]  |                   | [0.198, 0.630] | [0.055, 0.557]  |
| <b>R2</b> |                  |                | [−0.262, 0.226] |                   |                | [−0.252, 0.236] |
| <b>R3</b> |                  |                |                 |                   |                |                 |

Table 4: 95% confidence intervals for the correlation parameter  $\varrho$  obtained via Estimator 1. The table compares the intervals derived from the semi-parametric bootstrap ( $B = 10^3$  replicas) and the asymptotic estimating functions theory.

We then apply Estimator 2, which incorporates more information by substituting the parametric marginals with the ecdf of the strictly positive losses. The point estimates for the correlation parameter are detailed in Table 5, while the associated 95% CIs –again comparing bootstrap and asymptotic approaches– are reported in Table 6. Incorporating the empirical distribution of the loss severity slightly improves the precision of the calibration compared to Estimator 1. Nevertheless, the fundamental issue persists: the confidence intervals remain excessively wide. Even with Estimator 2, the interval widths for all pairs of risk classes exceed 0.30.

|           | <b>R1</b> | <b>R2</b> | <b>R3</b> |
|-----------|-----------|-----------|-----------|
| <b>R1</b> |           | 0.235     | 0.103     |
| <b>R2</b> |           |           | −0.060    |
| <b>R3</b> |           |           |           |

Table 5: Point estimates of the correlation parameter  $\hat{\varrho}$  obtained via Estimator 2 for each pair of operational risk categories.

These results are highly consequential. Despite utilizing the most comprehensive database available in the operational risk sector and observing a long time window of 180 months, the inherent sparsity of the data fundamentally limits the precision of purely statistical calibration. Indeed, the confidence intervals systematically span multiple discrete values of the regulatory correlation grid.

|           | Bootstrap |                |                 | Asymptotic |                |                 |
|-----------|-----------|----------------|-----------------|------------|----------------|-----------------|
|           | R1        | R2             | R3              | R1         | R2             | R3              |
| <b>R1</b> |           | [0.077, 0.395] | [-0.087, 0.293] |            | [0.066, 0.404] | [-0.095, 0.301] |
| <b>R2</b> |           |                | [-0.272, 0.129] |            |                | [-0.267, 0.147] |
| <b>R3</b> |           |                |                 |            |                |                 |

Table 6: 95% confidence intervals for the correlation parameter  $\varrho$  obtained via Estimator 2. The table compares the intervals derived from the semi-parametric bootstrap ( $B = 10^3$  replicas) and the asymptotic estimating functions theory.

### 4.3 Additional comments on Confidence Intervals

To systematically investigate the statistical uncertainty of our proposed methodologies, we analyze the theoretical width of the 95% asymptotic confidence intervals for the correlation parameter, denoted as  $\Delta\varrho$ . By holding the sample size fixed at  $N = 180$  (matching the length of our GGG dataset), we map how  $\Delta\varrho$  behaves across the entire domain of marginal occurrence probabilities,  $p_1$  and  $p_2$ , for various underlying true correlations  $\varrho \in \{0.00, 0.25, 0.50, 0.75\}$ .<sup>3</sup>

In Figure 1, we plot the combinations of  $p_1$  and  $p_2$  that allow the achievement of a  $\Delta\varrho$  lower than a given threshold, that corresponds to different colors; different shapes for such region are obtained depending on the true correlation  $\varrho$  of the copula. We observe that, for every value of  $\varrho$ , Estimator 1 achieves the best performance (i.e. minimal estimation uncertainty) when both time-series are half-full and half-empty ( $p_1$  and  $p_2$  around 0.5), but the width of the 95% CI is never below 0.25.

<sup>3</sup>We plot here only the case when  $\varrho$  is positive. The plots relative for negative  $\varrho$  are simply obtained by symmetry, by reflecting over the axis  $p_1 = 0.5$  or, equivalently, over the axis  $p_2 = 0.5$ .

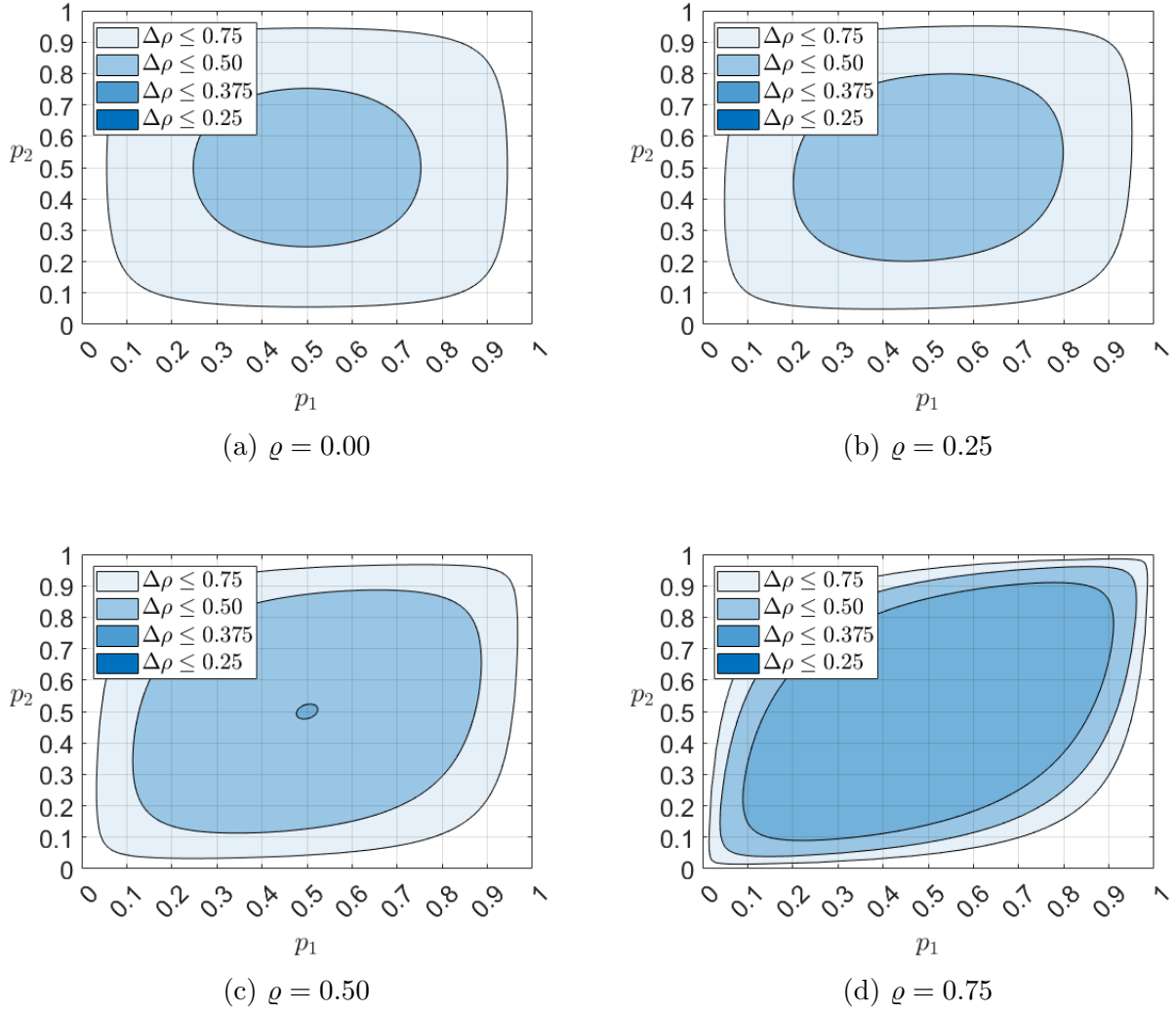


Figure 1: Estimation uncertainty for the semi-parametric Estimator 1 as a function of the true Gaussian copula correlation  $\rho$ . The shaded regions represent the combinations of marginal occurrence probabilities ( $p_1$  and  $p_2$ ) for which the width of the asymptotic 95% confidence interval, denoted as  $\Delta\rho$ , falls below a specific threshold. The intervals are derived using Corollary 3.3 for a sample size  $N = 180$ . Notably, the width  $\Delta\rho$  never drops below 0.25, regardless of the occurrence probabilities.

As in the case of Estimator 1, also for Estimator 2 we plot in Figure 2 the combinations of  $p_1$  and  $p_2$  that allow the achievement of a  $\Delta\rho$  lower than a given threshold, different colors correspond to different thresholds; different shapes for such region are obtained depending on the true correlation  $\rho$  of the copula. In this case, for larger correlation values, the width of the 95% CI is below 0.25 for large values of  $p_1$  and  $p_2$ .

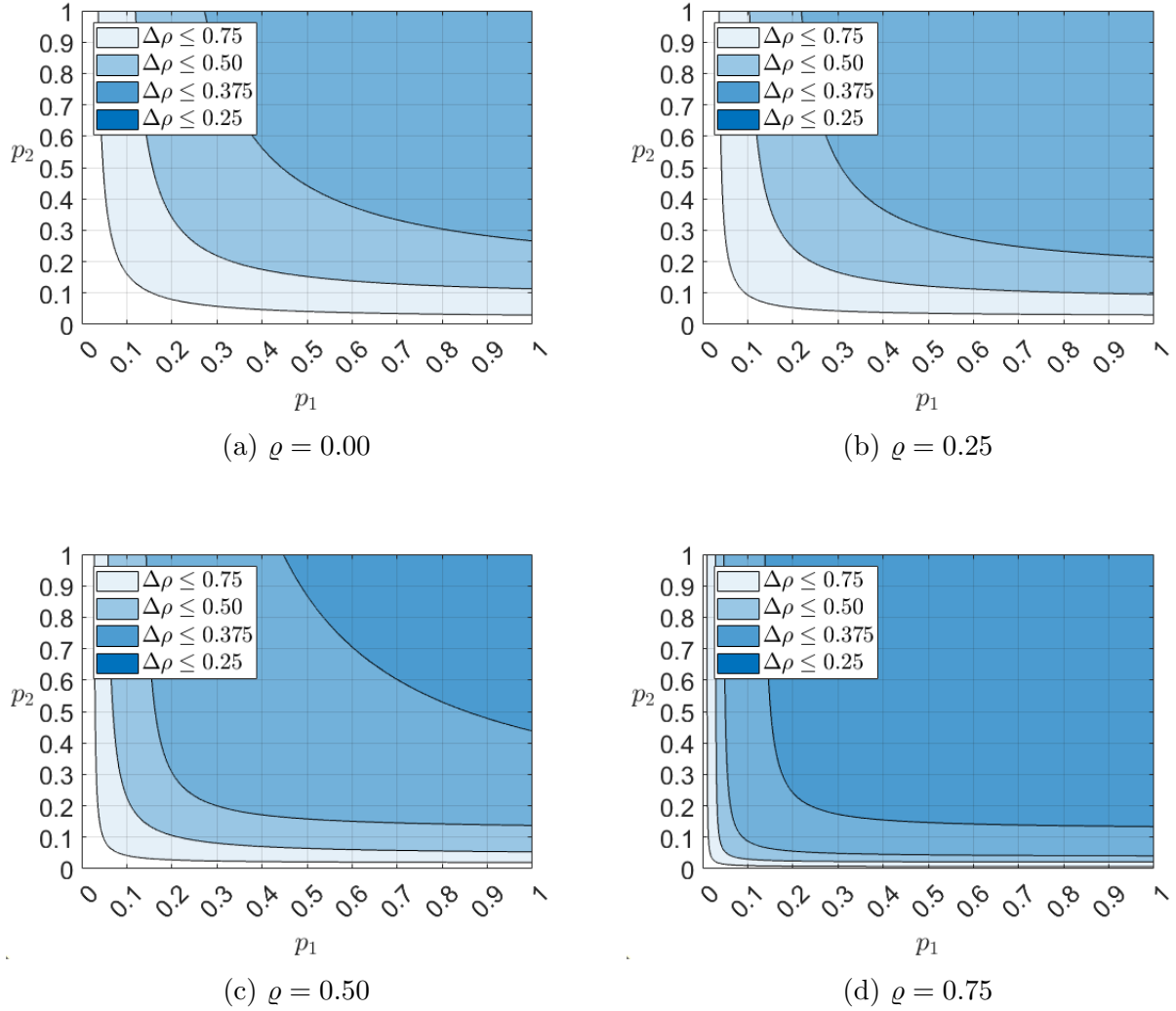


Figure 2: Estimation uncertainty for the semi-parametric Estimator 2 as a function of the true Gaussian copula correlation  $\rho$ . The shaded regions represent the combinations of marginal occurrence probabilities ( $p_1$  and  $p_2$ ) for which the width of the asymptotic 95% confidence interval,  $\Delta\rho$ , falls below a specific threshold. The intervals are derived using Corollary 3.6 for a sample size  $N = 180$ . While precision slightly improves for very dense time series ( $p_1, p_2 \rightarrow 1$ ), the interval width remains substantial across most of the domain.

## 5 Conclusions

In this work, we have introduced a simple yet effective semi-parametric framework for estimating operational risk correlations, developing two novel estimators that have improved accuracy and flexibility. We have also demonstrated a non-trivial application of asymptotic IFM theory to construct confidence intervals in a semi-parametric setting, ensuring more reliable statistical inference. Furthermore, we have conducted the first large-scale empirical analysis using the GGG dataset, covering operational losses in Western Europe and North America from 2008 to 2022, providing valuable insights into real-world loss dependencies.

Our findings have highlighted the inherent statistical uncertainty in correlation estimation for operational risk, showing that even with the largest available dataset in the insurance sector, confidence intervals have remained too wide for a parameter selection even within the simple set of quantized values utilized in the Standard model of Solvency II. This ambiguity indicates the necessity of integrating expert judgment into the process.

This study leads to a clear policy implication. Regulators and risk managers must recognize the limitations of purely data-driven methods and ensure that expert-based input is incorporated to enhance the robustness of internal models for operational risk while maintaining regulatory compliance.

## References

- Bandi, F.M., Kolokolov, A., Pirino, D., and Renò, R., 2020. Zeros, *Management Science*, 66 (8), 3466–3479.
- Baviera, R., Manzoni, P., and Massaria, M.D., 2026. Modelling dependence in sparse time series of insurance claims, *Unpublished manuscript, Available upon request*.
- Baviera, R. and Massaria, M.D., 2026. The additive bachelier model with an application to the oil option market in the covid period, *Journal of Computational and Applied Mathematics*, 487, 117741.
- Böcker, K. and Klüppelberg, C., 2010. Multivariate models for operational risk, *Quantitative Finance*, 10 (8), 855–869.
- Bolancé, C., Guillén, M., Gustafsson, J., and Nielsen, J.P., 2012. *Quantitative operational risk models*, CRC press.
- Brigo, D. and Morini, M., 2005. An empirically efficient analytical cascade calibration of the LIBOR Market Model based only on directly quoted swaptions data, *Available at SSRN 552581*.
- Casella, G. and Berger, R., 2024. *Statistical inference*, CRC Press.
- Cherubini, U., Luciano, E., and Vecchiato, W., 2004. *Copula methods in finance*, Wiley.
- Embrechts, P., 2002. Correlation and dependence in risk management: Properties and pitfalls, *Risk Management: Value at Risk and Beyond/Cambridge University Press*.
- European Parliament and Council of the European Union, 2009. Solvency II Directive (2009/138/EC), Available at: <https://eur-lex.europa.eu/eli/dir/2009/138/oj/eng>, Last accessed: 12/09/2025.

- Genest, C., Ghoudi, K., and Rivest, L.P., 1995. A semiparametric estimation procedure of dependence parameters in multivariate families of distributions, *Biometrika*, 82 (3), 543–552.
- GGG, 2026. GGG, accessed 12-03-2026.
- Godambe, V.P., 1991. *Estimating functions*, Oxford University Press.
- Grimmett, G. and Stirzaker, D., 2020. *Probability and random processes*, Oxford university press.
- Joe, H., 1997. *Multivariate models and multivariate dependence concepts*, CRC Press.
- Joe, H., 2005. Asymptotic efficiency of the two-stage estimation method for copula-based models, *Journal of multivariate Analysis*, 94 (2), 401–419.
- Kelliher, P., Acharyya, M., Couper, A., Grant, K., Maguire, E., Nicholas, P., Smerald, C., Stevenson, D., Thirlwell, J., and Cantle, N., 2017. Good practice guide to setting inputs for operational risk models, *British Actuarial Journal*, 22 (1), 68–108.
- Kelliher, P., Acharyya, M., Couper, A., Maguire, E., Nicholas, P., Pang, N., Smerald, C., Stevenson, D., Sullivan, J., and Teggin, P., 2020. Operational risk dependencies, *British Actuarial Journal*, 25, e5.
- Lu, T.T. and Shiou, S.H., 2002. Inverses of  $2 \times 2$  block matrices, *Computers & Mathematics with Applications*, 43 (1-2), 119–129.
- McNeil, A.J., Frey, R., and Embrechts, P., 2005. *Quantitative risk management: concepts, techniques and tools*, Princeton University Press.
- McNeil, A.J., Frey, R., and Embrechts, P., 2015. *Quantitative risk management: concepts, techniques and tools-revised edition*, Princeton university press.
- Oakes, D., 1994. Multivariate survival distributions, *Journal of Nonparametric Statistics*, 3 (3-4), 343–354.
- Tsukahara, H., 2005. Semiparametric estimation in copula models, *Canadian Journal of Statistics*, 33 (3), 357–375.
- Yip, K.C. and Yau, K.K., 2005. On modeling claim frequency data in general insurance with extra zeros, *Insurance: Mathematics and Economics*, 36 (2), 153–163.

# A Fundamentals of Estimating Functions Theory

In the theory of statistical inference, a fundamental problem is the estimation of unknown parameters that describe the dependence between observed data. Estimating functions theory plays a pivotal role in addressing this challenge by providing a framework to construct estimators in a flexible and general manner (see, e.g., Godambe 1991, Joe 1997). This framework forms the theoretical foundation for the semiparametric estimators proposed in Section 3.

Let  $\Theta \subseteq \mathbb{R}^k$  be an open space, and let  $\{\mathbf{X}^{(t)}\}_{t=1}^N$  be a sequence of  $N$  independent and identically distributed random vectors in  $\mathbb{R}^d$ , whose true distribution depends on an unknown parameter vector  $\boldsymbol{\theta} \in \Theta$ . We consider a vector-valued function  $\mathbf{g} : \mathbb{R}^d \times \Theta \rightarrow \mathbb{R}^k$ . Furthermore, let us define the expected Jacobian matrix  $D(\boldsymbol{\theta})$  and the covariance matrix  $M(\boldsymbol{\theta})$  associated with  $\mathbf{g}$  as follows:

$$D(\boldsymbol{\theta}) := -\mathbb{E}[J_{\mathbf{g}}(\mathbf{X}, \boldsymbol{\theta})] \in \mathbb{R}^{k \times k} \quad M(\boldsymbol{\theta}) := \mathbb{E}[\mathbf{g}(\mathbf{X}, \boldsymbol{\theta})\mathbf{g}(\mathbf{X}, \boldsymbol{\theta})^\top] \in \mathbb{R}^{k \times k}$$

where  $J_{\mathbf{g}}$  denotes the Jacobian matrix of the function  $\mathbf{g}$  with respect to  $\boldsymbol{\theta}$  (i.e.,  $(J_{\mathbf{g}})_{ij} = \partial g_i / \partial \theta_j$ ).

**Definition A.1.** (Vector-valued estimating function) A vector-valued estimating function  $\mathbf{g} : \mathbb{R}^d \times \Theta \rightarrow \mathbb{R}^k$  is a function satisfies the following conditions (cf. Godambe 1991, Section 1.6, pp.10-13):

$$\begin{cases} \mathbb{E}[\mathbf{g}(\mathbf{X}, \boldsymbol{\theta})] = \mathbf{0} \\ D(\boldsymbol{\theta}) \text{ is non-singular} \\ M(\boldsymbol{\theta}) \text{ is positive definite and finite} \end{cases} \quad \forall \boldsymbol{\theta} \in \Theta \quad (\text{A.1})$$

Given a finite sample of observations  $\{\mathbf{x}^{(t)}\}_{t=1}^N$ , the parameter estimator  $\hat{\boldsymbol{\theta}} \in \Theta$  is defined as the solution to the empirical estimating equations:

$$\sum_{t=1}^N \mathbf{g}(\mathbf{x}^{(t)}, \hat{\boldsymbol{\theta}}) = \mathbf{0}. \quad (\text{A.2})$$

For the asymptotic properties of estimating functions, the following Theorem holds (cf. Joe 1997, Section 10.1.1, pp.301-302).

**Theorem A.2.** Let  $\boldsymbol{\theta}$  be the true vector of parameters and let  $\hat{\boldsymbol{\theta}}$  be the estimator obtained by solving (A.2). Then:

i) the estimator  $\hat{\boldsymbol{\theta}}$  is consistent, i.e.

$$\hat{\boldsymbol{\theta}} \xrightarrow{a.s.} \boldsymbol{\theta},$$

ii) it is asymptotically Gaussian, i.e.

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{\text{law}} \mathcal{N}(\mathbf{0}, V(\boldsymbol{\theta})) \quad \text{as } N \rightarrow \infty,$$

where  $V(\boldsymbol{\theta})$  is the inverse Godambe information matrix:

$$V(\boldsymbol{\theta}) := D(\boldsymbol{\theta})^{-1} M(\boldsymbol{\theta}) (D(\boldsymbol{\theta})^{-1})^\top \in \mathbb{R}^{k \times k} \quad (\text{A.3})$$

## B Proofs

**Proof of Proposition 2.1.** The per-observation log-density (2.4) follows from a direct application of the copula definition to marginal cdfs featuring a probability atom at zero (similarly to what is done in Baviera *et al.* 2026, for the general case). The full sample log-likelihood (2.5) is straightforwardly obtained by summing the per-observation log-likelihoods across all  $N$  independent observations and grouping the terms according to the active sets  $\square$

**Lemma B.1.** *Given the  $D$  matrix in (3.4), its inverse is*

$$D^{-1} := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -d_{31}/d_{33} & -d_{32}/d_{33} & -1/d_{33} \end{bmatrix} \quad (\text{B.1})$$

*Proof.* Straightforward using a well-known result of block matrices inversion (cf., e.g., Lu and Shiu 2002, Theorem 2.1)  $\square$

**Proof of Proposition 3.1.** Indeed, we can easily deduce  $\hat{p}_1, \hat{p}_2, \hat{\rho}$  from the form for the vector of estimates (3.1). We first obtain the estimators for parameters for the two risk classes separately (i.e.  $\hat{p}_1, \hat{p}_2$ ), then we obtain the estimator for correlation  $\hat{\rho}$  given the other two parameters  $\square$

**Proof of Proposition 3.2.** We first show that the vector of estimating functions (3.1) satisfies the set of hypotheses A.1.

It is immediate to prove that

$$\mathbb{E}[g_i(p_1, p_2, \rho)] = 0 \quad \forall [p_1, p_2, \rho]^\top \in [0, 1] \times [0, 1] \times (-1, 1) \quad i = 1, 2, 3.$$

We can prove that the expectation of the Jacobian matrix is equal to the following matrix

$$D := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ d_{31} & d_{32} & d_{33} \end{bmatrix} \quad \text{where} \quad \begin{cases} d_{31} := \Phi \left( \frac{1}{\sqrt{1-\rho^2}} [\Phi^{-1}(1-p_2) - \rho \Phi^{-1}(1-p_1)] \right) \\ d_{32} := \Phi \left( \frac{1}{\sqrt{1-\rho^2}} [\Phi^{-1}(1-p_1) - \rho \Phi^{-1}(1-p_2)] \right) \\ d_{33} := \varphi_2(\Phi^{-1}(1-p_1), \Phi^{-1}(1-p_2); \rho) \end{cases}, \quad (\text{B.2})$$

that is non-singular  $\forall [p_1, p_2, \rho]^\top \in [0, 1] \times [0, 1] \times (-1, 1)$ , because  $d_{33}$  is always greater than zero. It remains to prove that the matrix

$$\mathbb{E} [\mathbf{g}(p_1, p_2, \rho) \mathbf{g}(p_1, p_2, \rho)^\top]$$

is positive definite and finite  $\forall [p_1, p_2, \rho]^\top \in [0, 1] \times [0, 1] \times (-1, 1)$ , the second condition is satisfied thanks to the fact that the estimating function are always bounded. The matrix is positive definite if, by definition,

$$\begin{cases} \forall \mathbf{z} \in \mathbb{R}^3 & \mathbf{z}^\top \mathbb{E} [\mathbf{g}(p_1, p_2, \rho) \mathbf{g}(p_1, p_2, \rho)^\top] \mathbf{z} \geq 0 \\ \mathbf{z}^\top \mathbb{E} [\mathbf{g}(p_1, p_2, \rho) \mathbf{g}(p_1, p_2, \rho)^\top] \mathbf{z} = 0 & \Leftrightarrow \mathbf{z} = \mathbf{0} \end{cases}.$$

We prove that

$$\mathbf{z}^\top \mathbb{E} [\mathbf{g}(p_1, p_2, \rho) \mathbf{g}(p_1, p_2, \rho)^\top] \mathbf{z} = \mathbb{E} \left[ \left( \mathbf{z}^\top \mathbf{g}(p_1, p_2, \rho) \right)^2 \right] \geq 0 \quad \forall \mathbf{z} \in \mathbb{R}^3 \quad \forall p_1, p_2, \rho,$$

and that

$$\mathbb{E} \left[ \left( \mathbf{z}^\top \mathbf{g}(p_1, p_2, \rho) \right)^2 \right] = 0 \quad \forall \mathbf{z} \in \mathbb{R}^3 \quad \Leftrightarrow \quad \mathbf{z}^\top \mathbf{g}(p_1, p_2, \rho) = 0$$

for any  $p_1, p_2, \rho$ . By straightforward calculations, the only solution is  $\mathbf{z} = \mathbf{0}$ . Thus, the matrix

$$\mathbb{E} [\mathbf{g}(p_1, p_2, \rho) \mathbf{g}(p_1, p_2, \rho)^\top]$$

is positive definite  $\forall p_1, p_2, \rho \in [0, 1] \times [0, 1] \times (-1, 1)$ .

We exploit Theorem A.2. We need to compute the matrices  $M$  and  $D$ , defined as

$$(M)_{ij} = \mathbb{E}[g_i(\boldsymbol{\theta}) g_j(\boldsymbol{\theta})] \quad \text{and} \quad (D)_{ij} = -\mathbb{E}[\partial g_i / \partial \theta_j],$$

with  $\boldsymbol{\theta} = [p_1, p_2, \rho]$  which are obtained by straightforward calculations □

**Proof of Corollary 3.3** Thanks to Proposition 3.2, the asymptotic distribution of the estimator  $\hat{\rho}$  is

$$\hat{\rho} \sim \mathcal{N}(\rho, \Sigma),$$

where the variance  $\Sigma$  corresponds to the element in third row and third column of the inverse Godambe information matrix

$$\frac{1}{N} D^{-1} M (D^{-1})^\top.$$

Thus

$$\Sigma = \frac{1}{N} v^\top M v,$$

where  $v^\top$  is the third row of the matrix  $D^{-1}$ , whose expression is given by Lemma B.1 □

**Proof of Proposition 3.4.** As for the other semi-parametric estimator, we can easily deduce  $[\hat{p}_1, \hat{p}_2, \hat{\rho}]^\top$  from the form for the vector of estimates (3.6). We first obtain the estimators for parameters for the two risk classes separately (i.e.  $\hat{p}_1, \hat{p}_2$ ), then we obtain the estimator for correlation  $\hat{\rho}$  given the other two parameters.  $\square$

**Proof of Proposition 3.5.** This result follows from Theorem 1, p.363 in the work of Tsukahara (2005), with the vector of functions

$$[g_1(p_1, p_2, \rho), g_2(p_1, p_2, \rho), g_\rho(p_1, p_2, \rho)]^\top.$$

The matrices  $M$  and  $D$  are obtained by straightforward calculations  $\square$

**Proof of Corollary 3.6.** The proof is analogous to that of Corollary 3.3  $\square$

**Proof of Proposition 3.7.** The result follows directly from Proposition 3.2 in Baviera *et al.* (2026), applied to the derivative of the log-likelihood with respect to  $\rho$ .<sup>4</sup> The fundamental difference in our semi-parametric framework is that the parametric cumulative distribution functions  $F_i(x_i; p_i)$  are replaced by the empirical cumulative distribution functions (ecdf)  $u_i$ . As the occurrence probabilities  $p_1$  and  $p_2$  approach 1, the indicator functions associated with the probability masses at zero vanish. Consequently, the estimating equation smoothly converges almost surely to the score of a standard bivariate Gaussian copula calibrated on continuous marginals, yielding the limiting function  $\bar{h}(\rho)$ .  $\square$

**Proof of Theorem 3.8.** The convergence of the estimator follows the same logical steps detailed in Theorem 3.3 of Baviera *et al.* (2026). Since the empirical estimating function converges locally uniformly almost surely to  $\bar{h}(\rho)$  as  $p_1, p_2 \rightarrow 1$  (as established in Proposition 3.8), the root  $\hat{\rho}$  of the finite-sample score equation converges almost surely to  $\bar{\rho}$ , which is the unique root of the limiting continuous equation  $\bar{h}(\bar{\rho}) = 0$ .  $\square$

---

<sup>4</sup>The interchange between the limit as  $p_1, p_2 \rightarrow 1$  and the differentiation with respect to  $\rho$  is mathematically justified by the local uniform convergence of the log-likelihood (cf. Baviera *et al.* 2026, Prop.C.2, App.C) and its derivatives on any compact subset of the parameter space  $(-1, 1)$ , as detailed in the fully parametric analysis.

| Symbol                                                  | Description                                                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| $\mathbf{Y} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ | vector of rvs $\mathbf{Y}$ distributed as a normal with mean $\boldsymbol{\mu}$ and covariance matrix $\Sigma$ |
| $\varphi(\bullet), \Phi(\bullet)$                       | pdf and cdf of a standard normal rv                                                                            |
| $\mathcal{M}(x)$                                        | $\varphi(x)/\Phi(-x)$ , inverse Mill's ratio                                                                   |
| $\varphi_2(x, y; \rho)$                                 | pdf of a bivariate standard normal distribution with correlation $\rho \in (-1, 1)$                            |
| $\Phi_2(x, y; \rho)$                                    | cdf of a bivariate standard normal distribution with correlation $\rho \in (-1, 1)$                            |
| $C(u, v; \rho)$                                         | bivariate Gaussian copula with parameter $\rho \in (-1, 1)$                                                    |
| $c(u, v; \rho)$                                         | bivariate Gaussian copula density                                                                              |
| $\overline{C}(u, v; \rho)$                              | $C(1 - u, 1 - v; \rho)$ : bivariate complementary Gaussian copula function                                     |
| $f(\bullet), F(\bullet)$                                | pdf and cdf of 1-d Comb-Bernoulli                                                                              |
| $\psi(\bullet), \Psi(\bullet)$                          | generic pdf and cdf of the marginal distribution for the losses                                                |
| $\ell(p_1, p_2, \rho \mid x_1, x_2)$                    | per-observation log-likelihood                                                                                 |
| $\mathcal{L}(p_1, p_2, \rho \mid \mathcal{X})$          | log-likelihood given the time-series $\mathcal{X}$                                                             |
| $N$                                                     | time-series length                                                                                             |
| $n^\parallel, n_0$                                      | number of co-jumps and no-jumps                                                                                |
| $n_1, n_2$                                              | number of non-null losses in the first (second) risk class                                                     |
| $X_1, X_2$                                              | loss in the first (second) risk class                                                                          |
| $\mathcal{X}$                                           | time series of operational losses                                                                              |
| $p_1, p_2$                                              | probability of non-null loss in the two risk classes                                                           |
| $p^\parallel, p_0$                                      | probabilities of a co-jump and a no-jumps                                                                      |
| $\rho$                                                  | Gaussian copula correlation                                                                                    |
| $\hat{\rho}$                                            | estimated value of the Gaussian copula correlation                                                             |
| $\bar{\rho}$                                            | estimated value of the Gaussian copula correlation when $p_1, p_2 \rightarrow 1$                               |
| $t$                                                     | time-series index                                                                                              |
| $\xrightarrow{\text{a.s.}}$                             | almost surely convergence                                                                                      |
| $\xrightarrow{\text{law}}$                              | convergence in law                                                                                             |
| $\delta_0(x)$                                           | Dirac delta function, centered in 0                                                                            |
| $\mathcal{C}^k$                                         | class of $k$ -times differentiable functions, with $k$ -th continuous derivative                               |

| Shorthand | Description                                |
|-----------|--------------------------------------------|
| a.s.      | almost surely                              |
| ecdf      | empirical cumulative distribution function |
| cdf       | cumulative distribution function           |
| CI        | Confidence Interval                        |
| ecdf      | empirical cumulative distribution function |
| IFM       | Inference Functions for Margins method     |
| i.i.d.    | independent and identically distributed    |
| pdf       | probability density function               |
| rv        | random variable                            |
| st.n.     | standard normal (random variable)          |
| wrt       | with respect to                            |